

PHENOMENAL ELECTRONIC DESIGN SOFTWARE To Impress The Engineer In You

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Xcos

Distributed along Scilab, Xcos is a tool used to model and simulate hybrid dynamic systems. The user can model discrete as well as continuous, causal as well as acausal models on it along with being able to design, load, save, compile and simulate. Xcos makes the most-commonly-used blocks such as signal processing, electrical systems, thermohydraulic systems, mathematical operations and much more. There are also ways to import user-defined blocks that can be easily added to Xcos framework. This tool is flexible enough to handle all that.

OpenModelica

OpenModelica is an open source Modelica based modelling and simulation environment intended for industrial and academic usage. The compiler used by OpenModelica converts objects from base Mod- elica code to C code. The user can use APIs that are freely available to interact with the system through Advanced Interactive OpenModelica Compiler. This is the crucial part of the package as it acts like the engine of the simulator.

FreeCAD PCB

Printed circuit board (PCB) software are designed to enable rapid develop- ment of electronic schematics and PCB layouts. This enables beginners as well as advanced users to quickly move their ideas into physical PCBs, aiding in faster project development. FreeCAD-PCB allows users to import PCB boards to FreeCAD. The software supports many layers along with op- tions to choose from various colours,

Some popular resources

KompoZer. KompoZer lets users edit Web pages and is a complete Web authoring system. With a robust file-management system, KompoZer has a user interface that is also ideal for non-technical computer users who can fulfil their wish to create professionally-attractive Web pages without prior knowledge of HTML or complex codes.

Chromium Portable. Chromium Portable is a custom-built version of Chromium Web browser. The browser is completely open and optimised for portable use. Reviews suggest that it runs seamlessly on Linux based platforms but is a little tough or sluggish on Windows. Chromium Portable leaves no personal information behind on any device like its well-known cousin Chrome browser. The software is based on Google Chrome development release channel, and is updated at least twice per month.

Ubuntu OS. Ubuntu could be one of the most popular open source Debian based OS anybody could come across. Perfect for installation on personal computers, smartphones and network servers, Ubuntu 16.04 comes with the latest versions of apps such as Firefox 45, Thunderbird 38, Files 3.14.2, Videos 3.18, LibreOffice 5.1, GNOME Terminal 3.18, Eye of GNOME 3.18, Rhythmbox 3.3, Shotwell 0.22 and many more.

There is something called Snap packages in the latest version of Ubuntu. It is a new way devised by the developer team of Canonical to package and deliver applications to Ubuntu users in order to deplete dependency factors. Snaps could be called a bundle with all dependencies and libraries, which assist users in installing the latest version of apps without going through any hassles of compatibility. And, with this version of Ubuntu, they can see that GNOME software is becoming the default software management and it facilitates upgrades of OS and the firmware.

names of layers and transparency. It also facilitates import of various Initial Graphics Exchange Specification (IGES) models with colours.

ScadaBR

ScadaBR is a Supervisory Control and Data Acquisition (SCADA) oper- ating system (OS) with applications in process control and automation. It is being developed and distributed using the open source model. This

software is aimed at supporting engineers with multiple use cases in the areas of industrial process automation, distribution networks, automation, land and residential applications. ScadaBR supports over 20 protocols of communication and is compatible with hardware of vari- ous manufacturing sectors.

Some interesting features that extend the functionality of this soft- ware are graphs and reporting, crea- tion of edition screens (views/HMI), access control data and commands, revenue programming (scripting), alert alarms and events manage- ment, among others.

FreeRTOS real-time kernel

A real-time operating system (RTOS) is optimised for applications in embedded/real-time environments. The primary objective here is to en- sure a prompt and timely response to events. FreeRTOS could become your one-stop-shop solution in the universe of RTOSes while dealing with microcontrollers and small microprocessors. The OS blends in with the hardware and is suited for any imaginable applications right from toys to aircrafts.

Each task in FreeRTOS could be assigned priority while the queuing methodology in the OS ensures that independent tasks are running and waiting tasks are equally managed. The real-time kernel featured in this OS is also responsible for task man- agement, and inter-task communi- cation and synchronisation, and it seamlessly handles such responsi- bilities in real-time. **EFY**

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